**Code:**

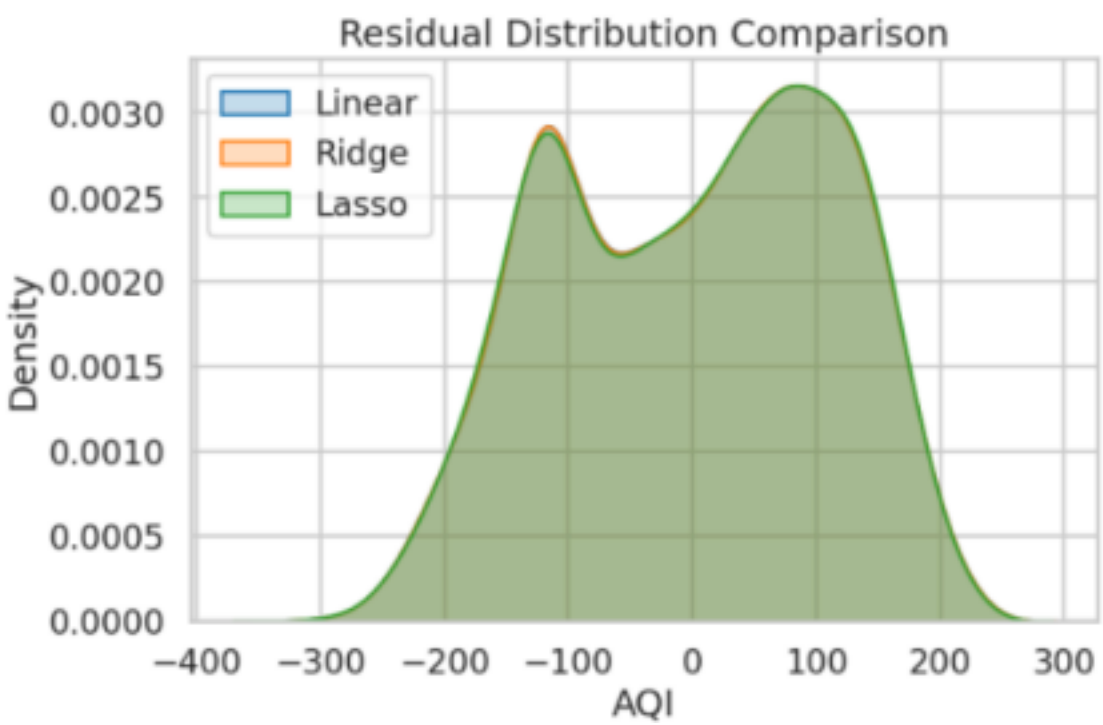
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split, GridSearchCV from sklearn.linear_model
import LinearRegression, Ridge, Lasso from sklearn.preprocessing import StandardScaler
from sklearn.metrics import mean_squared_error, r2_score
sns.set_style("whitegrid")
sns.set_context("talk")
from google.colab import files
print("Upload your CSV dataset")
uploaded = files.upload()
file_name = list(uploaded.keys())[0]
try:
df = pd.read_csv(file_name)
except:
df = pd.read_csv(file_name, sep=';', engine='python')
print("\n ✅ Dataset Loaded")
print("Original Shape:", df.shape)
df = df.select_dtypes(include=np.number) df = df.dropna()
print("After Cleaning Shape:", df.shape)
target_column = df.columns[-1]
print("Target Column:", target_column)
X = df.drop(target_column, axis=1)
y = df[target_column]
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
X_train, X_test, y_train, y_test = train_test_split( X_scaled, y, test_size=0.2, random_state=42 )
lr = LinearRegression()
lr.fit(X_train, y_train)
y_pred_lr = lr.predict(X_test)
ridge = Ridge()
param_ridge = {'alpha':[0.001,0.01,0.1,1,10,100]} grid_ridge = GridSearchCV(ridge,
param_ridge, cv=5) grid_ridge.fit(X_train, y_train)
best_ridge = grid_ridge.best_estimator_
y_pred_ridge = best_ridge.predict(X_test)
lasso = Lasso(max_iter=10000)
param_lasso = {'alpha':[0.001,0.01,0.1,1,10]} grid_lasso = GridSearchCV(lasso, param_lasso,
cv=5) grid_lasso.fit(X_train, y_train)
best_lasso = grid_lasso.best_estimator_
y_pred_lasso = best_lasso.predict(X_test)
def evaluate(name, y_true, y_pred):
print(f"\n{name}")
```

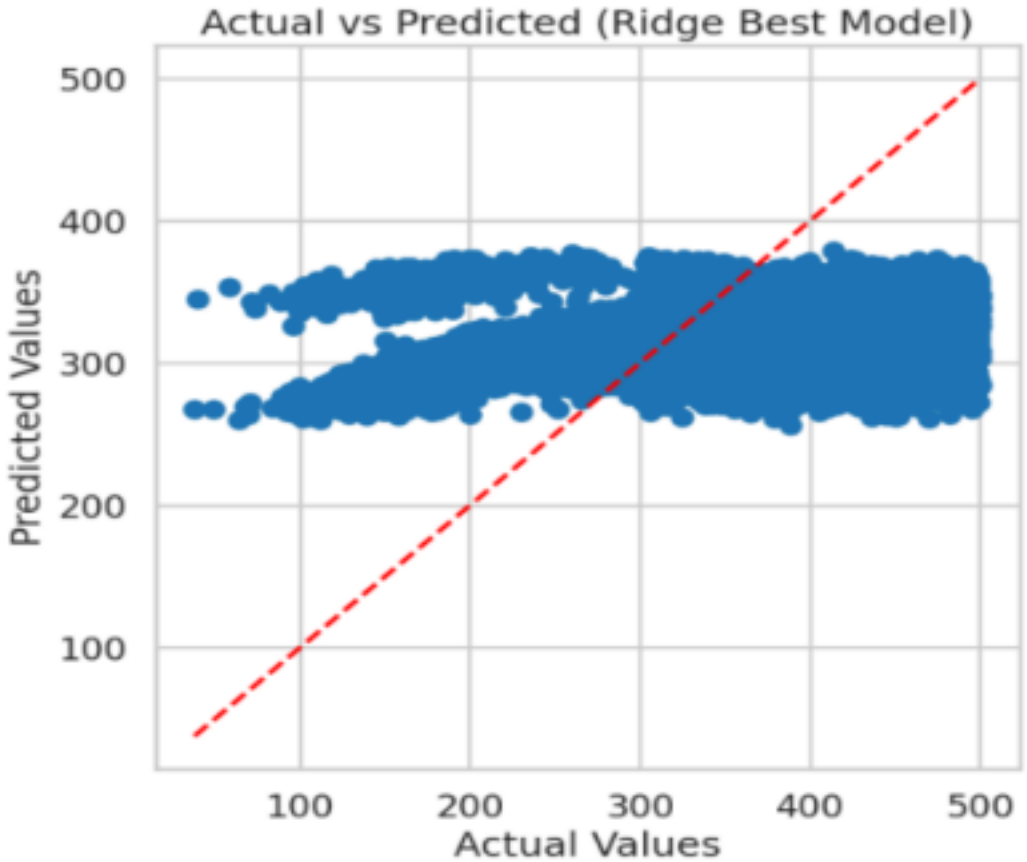
```
print("R2 Score :", r2_score(y_true, y_pred))
print("MSE :", mean_squared_error(y_true, y_pred)) print("RMSE :",
np.sqrt(mean_squared_error(y_true, y_pred)))
evaluate("Linear Regression", y_test, y_pred_lr)
evaluate("Best Ridge Regression", y_test, y_pred_ridge) evaluate("Best Lasso Regression",
y_test, y_pred_lasso)
print("\nBest Ridge Alpha:", grid_ridge.best_params_) print("Best Lasso Alpha:",
grid_lasso.best_params_)
models = ["Linear", "Ridge", "Lasso"]
r2_scores = [
r2_score(y_test, y_pred_lr),
r2_score(y_test, y_pred_ridge),
r2_score(y_test, y_pred_lasso)
]
plt.figure(figsize=(8,5))
sns.barplot(x=models, y=r2_scores)
plt.title("R2 Score Comparison")
plt.show()
residual_lr = y_test - y_pred_lr
residual_ridge = y_test - y_pred_ridge
residual_lasso = y_test - y_pred_lasso
plt.figure(figsize=(8,5))
sns.kdeplot(residual_lr, label="Linear", fill=True)
sns.kdeplot(residual_ridge, label="Ridge", fill=True) sns.kdeplot(residual_lasso, label="Lasso",
fill=True) plt.title("Residual Distribution Comparison")
plt.legend()
plt.show()
plt.figure(figsize=(7,7))
plt.scatter(y_test, y_pred_ridge)
plt.plot([y_test.min(), y_test.max()],
[y_test.min(), y_test.max()],
'r--')
plt.xlabel("Actual Values")
plt.ylabel("Predicted Values")
plt.title("Actual vs Predicted (Ridge Best Model)") plt.show()
errors = pd.DataFrame({
"Linear": residual_lr,
"Ridge": residual_ridge,
"Lasso": residual_lasso
})
plt.figure(figsize=(8,5))
sns.boxplot(data=errors)
plt.title("Error Distribution Comparison")
plt.show()
```

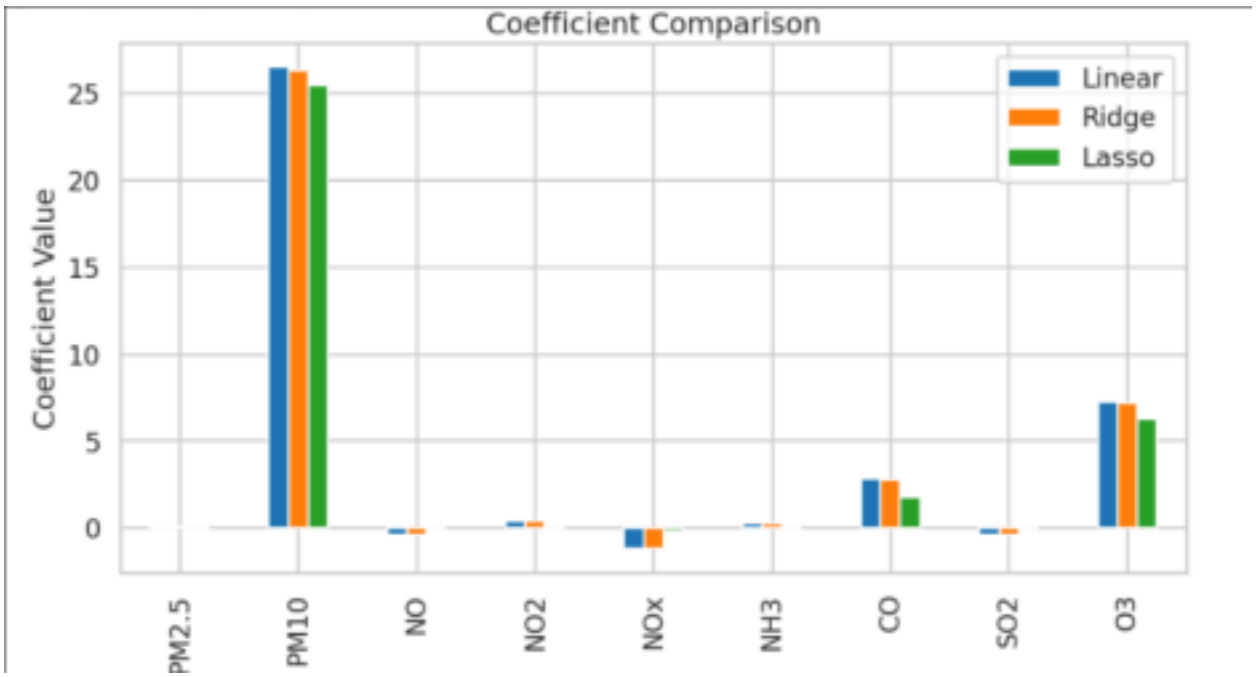
```
coef_df = pd.DataFrame({
"Feature": X.columns,
"Linear": lr.coef_,
"Ridge": best_ridge.coef_,
"Lasso": best_lasso.coef_
})
coef_df.set_index("Feature").plot(kind="bar", figsize=(12,6)) plt.title("Coefficient Comparison")
plt.ylabel("Coefficient Value")
plt.show()
importance = pd.DataFrame({
"Feature": X.columns,
"Importance": np.abs(best_ridge.coef_)
}).sort_values("Importance", ascending=False)
plt.figure(figsize=(10,6))
sns.barplot(data=importance.head(10),
x="Importance", y="Feature")
plt.title("Top 10 Important Features (Ridge)")
plt.show()
alphas = param_ridge['alpha']
ridge_scores = []
for a in alphas:
model = Ridge(alpha=a)
model.fit(X_train, y_train)
pred = model.predict(X_test)
ridge_scores.append(r2_score(y_test, pred))
plt.figure(figsize=(8,5))
plt.plot(alphas, ridge_scores, marker='o')
plt.xscale("log")
plt.xlabel("Alpha")
plt.ylabel("R2 Score")
plt.title("Ridge Hyperparameter Tuning Curve")
plt.show()
zero_count = np.sum(best_lasso.coef_ == 0)
print("\nNumber of features eliminated by Lasso:", zero_count) print("\n ✅Experiment
Completed Successfully")
```

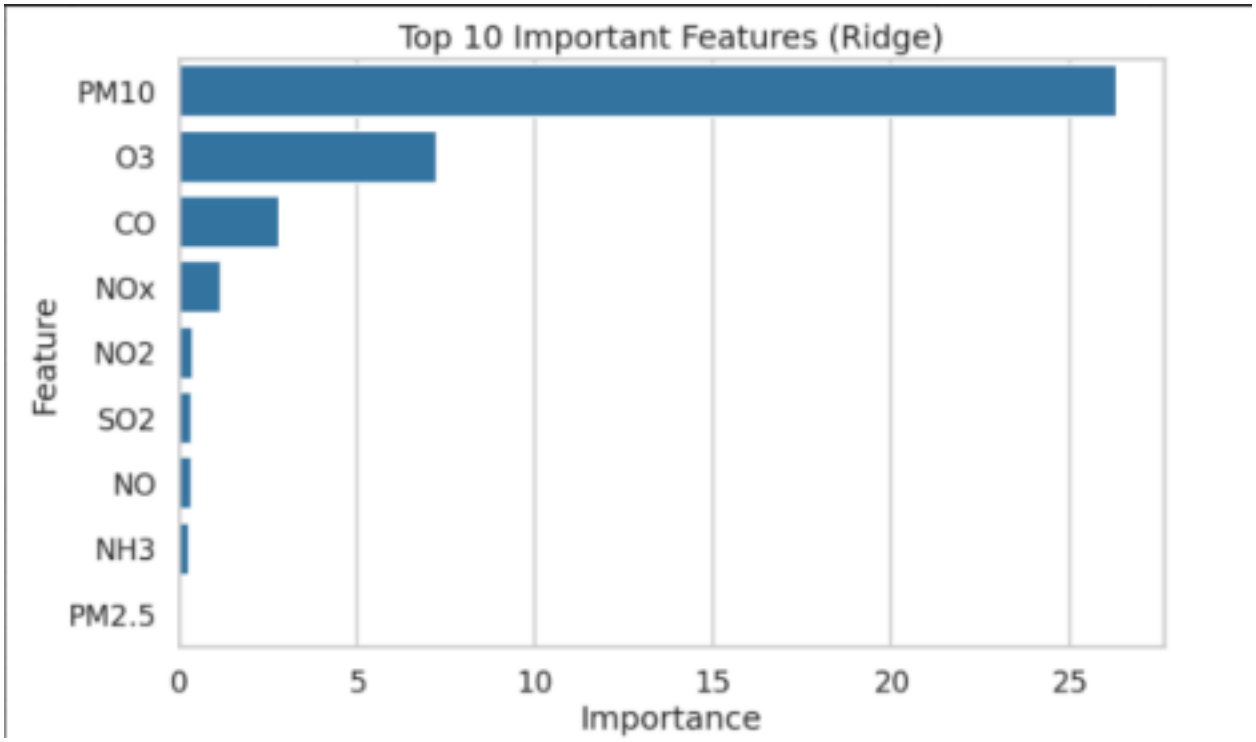
**OUTPUT:**

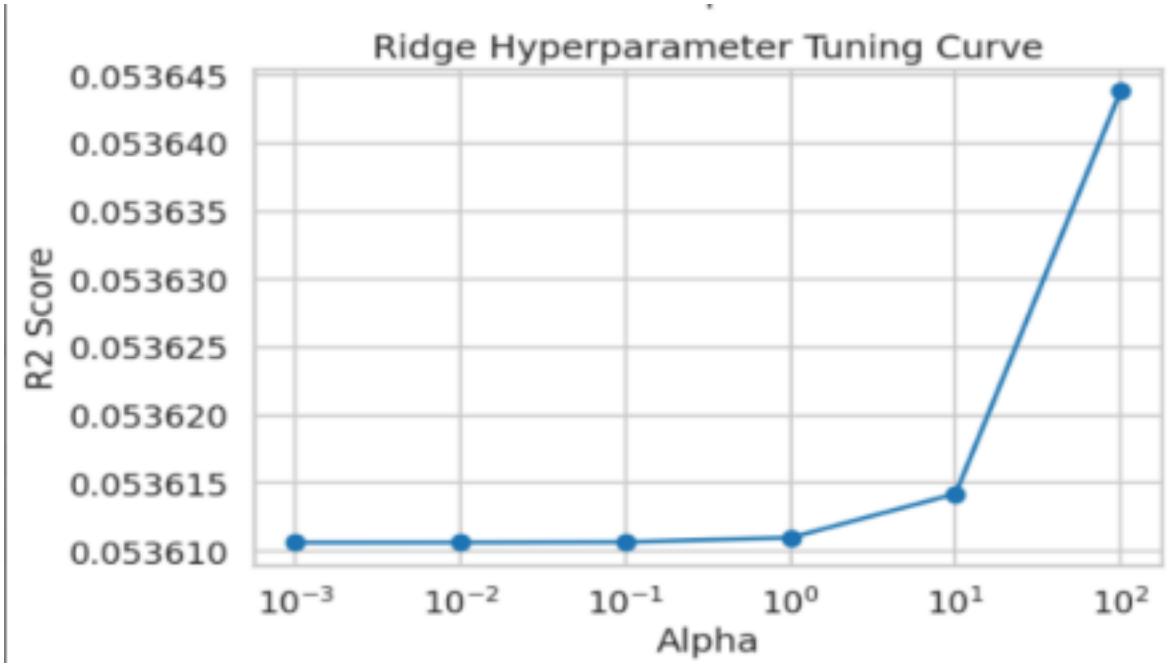
```
Upload your CSV dataset
Choose Files  Air_quality_data.csv
Air_quality_data.csv(text/csv) - 1494181 bytes, last modified: 2/16/2026 - 100% done
Saving Air_quality_data.csv to Air_quality_data (2).csv

✅ Dataset Loaded
Original Shape: (18265, 13)
After Cleaning Shape: (18265, 10)
Target Column: AQI

Linear Regression
R2 Score : 0.053610632587749674
MSE      : 12606.61964732814
RMSE     : 112.27920398421134

Best Ridge Regression
R2 Score : 0.05364384351460827
MSE      : 12606.177252752019
RMSE     : 112.27723390230105

Best Lasso Regression
R2 Score : 0.05337494980178081
MSE      : 12609.759119666312
RMSE     : 112.29318376315774

Best Ridge Alpha: {'alpha': 100}
Best Lasso Alpha: {'alpha': 1}
```




Actual vs Predicted (Ridge Best Model)
Predicted Values
Actual Values
500
400
300
200
100
100
200
300
400
500


Coefficient Comparison
Coefficient Value
Linear
Ridge
Lasso
25
20
15
10
5
0
PM2.5
PM10
NO
NO2
NOx
NH3
CO
SO2
O3


Top 10 Important Features (Ridge)
PM10
O3
CO
NOx
NO2
SO2
NO
NH3
PM2.5
Feature
0
5
10
15
20
25
Importance


Ridge Hyperparameter Tuning Curve
0.053645
0.053640
0.053635
0.053630
0.053625
0.053620
0.053615
0.053610
R2 Score
$10^{-3}$
$10^{-2}$
$10^{-1}$
$10^{0}$
$10^{1}$
$10^{2}$
Alpha